

Definition 0.1 (Kullback–Leibler Divergence): Let P and Q be probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. The Kullback–Leibler divergence (or relative entropy) of P with respect to Q is defined as:

$$D_{\text{KL}}(P\|Q) = \begin{cases} \int_{\mathcal{X}} \log\left(\frac{dP}{dQ}\right) dP & \text{if } P \ll Q \\ +\infty & \text{otherwise} \end{cases}$$

Using the density $p = \frac{dP}{d\mu}$ and $q = \frac{dQ}{d\mu}$ relative to a common reference measure μ , this is:

$$D_{\text{KL}}(P\|Q) = \int_{\mathcal{X}} p(x) \log \frac{p(x)}{q(x)} d\mu(x),$$

whenever $P \ll Q$ and $p \log(p/q)$ is μ -integrable.

Remark 0.2: KL divergence is a special case of f -divergence with $f(t) = t \log t$. Its second-order expansion around $P = Q$ induces the Fisher information metric, which underlies the local geometry of statistical models.

Remark 0.3: Gibbs' Inequality states that $D_{\text{KL}}(P\|Q) \geq 0$, with equality if and only if $P = Q$ almost everywhere. This follows from Jensen's inequality applied to the convex function $t \log t$ (or equivalently to $-\log t$ under P).

Remark 0.4: $D_{\text{KL}}(P\|Q)$ is not a true metric as it lacks symmetry ($D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$) and does not satisfy the triangle inequality. In Bayesian inference, the asymmetry distinguishes between the "projection" of a complex posterior onto a simpler family (variational inference) versus the reverse.

Remark 0.5: The KL divergence is invariant under coordinate transformations (reparameterizations). If $T : \mathcal{X} \rightarrow \mathcal{Y}$ is a measure-preserving isomorphism, then $D_{\text{KL}}(P\|Q) = D_{\text{KL}}(T_*P\|T_*Q)$, where T_* is the push-forward measure.

0.0.1 See also

[Mutual Information]

0.0.2 References